LP-based branch-and-bound (2/2)

Outline

Worked examples on the car example

Typical mistakes

Extensions

In summary

Where we were

- In the first part, the B&B algorithm: branching on a fractional integer variable, $x_j \leq \lfloor \tilde{x}_j \rfloor$ and $x_j \geq \lfloor \tilde{x}_j \rfloor + 1$
- Pruning by infeasibility, by integrality and by bound, with the bound of the node $\mathrm{UB}(\mathrm{LP}_k) = \lfloor z(\mathrm{LP}_k) \rfloor$ when all the variables and costs are integer
- LB from the incumbent solution, $\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k)$, and the skip rule
- Each child is re-optimized with the dual simplex algorithm; the strategies choose the branching variable and the next node
- **Now**: the complete executions of the algorithm on the car example

Worked examples on the car example

Introduction: Worked examples on the car example

- We solve the car example with the B&B algorithm, step by step
- Two executions with different selection strategies give the same optimal solution with trees of different size
- A third execution starts from a heuristic solution, and shows how much a good lower bound helps

Two executions of the B&B algorithm

- We now solve the car example with two executions of the B&B algorithm, which differ only in the selection strategies
- Since all the variables and all the costs are integer, the bound of a node is

$$\mathrm{UB}(\mathrm{LP}_k) = \lfloor z(\mathrm{LP}_k) \rfloor$$

- In the first execution every node is re-optimized with the dual simplex algorithm, and we report all the dictionaries
- For each node a figure shows its feasible region

Before reading the solution

Before reading the solution, try to predict
which variable is selected at the root,

and which node is extracted
at the second iteration.

Example: execution A, the rules

We execute the B&B algorithm on the car example with:

1. the smallest-index rule (lexicographic) as branching-variable selection strategy;
2. the LIFO rule as branching-node selection strategy;
3. the down-branching node is solved first, then the up-branching node;
4. the second child is not solved when the skip rule applies;
5. the lower bound is initialized to $\mathrm{LB} = -\infty$ (no heuristic algorithm is executed).

Example: execution A, initialization (1/3)

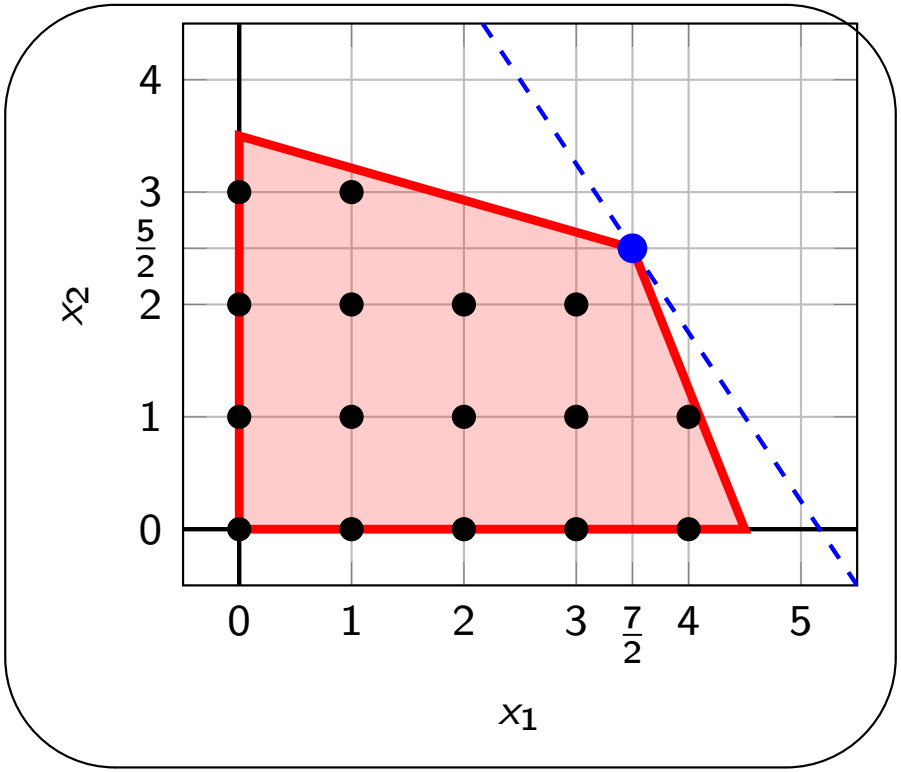
- The LP_1 problem is the LP relaxation of the car example
- The simplex algorithm with Bland's rule, starting from the basis of the slack variables, performs 2 pivoting operations (x_1 enters and x_3 leaves; x_2 enters and x_4 leaves) and ends with the optimal dictionary:

$$\begin{cases} x_1 = \frac{7}{2} - \frac{7}{62} x_3 + \frac{1}{31} x_4 \\ x_2 = \frac{5}{2} + \frac{1}{31} x_3 - \frac{5}{62} x_4 \\ z = \frac{31}{2} - \frac{17}{62} x_3 - \frac{2}{31} x_4 \end{cases}$$

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_1)$ is:

$$\tilde{x}_1 = \frac{7}{2}, \quad \tilde{x}_2 = \frac{5}{2}, \qquad z(\mathrm{LP}_1) = \frac{31}{2}.$$

Example: execution A, initialization (2/3)



Node 1: the feasible region of LP_1, the integer points, the optimal solution of LP_1 (blue) and the objective line through it (dashed).

Example: execution A, initialization (3/3)

- The solution is fractional and

$$\lfloor z(\mathrm{LP}_1) \rfloor = \left\lfloor \frac{31}{2} \right\rfloor = 15 > \mathrm{LB} = -\infty :$$

  the LP_1 problem is inserted in the list L

- The list is $L = \{\mathrm{LP}_1\}$, and the upper bound is the largest bound of the nodes in the list:

$$\underbrace{-\infty}_{\mathrm{LB}} \leq z(\mathrm{ILP}) \leq \underbrace{15}_{\mathrm{UB}} \quad \left(\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k) = 15 \right).$$

Example: execution A, iteration 1

- The list is $L = \{\mathrm{LP}_1\}$: the LP_1 problem is extracted and removed from L
- In $\tilde{\boldsymbol{x}}(\mathrm{LP}_1)$ we have $\tilde{x}_1 = \frac{7}{2}$ and $\tilde{x}_2 = \frac{5}{2}$, both fractional; by the smallest-index rule the branching variable is x_1
- The branching constraints are:

$$x_1 \leq \underbrace{\left\lfloor \frac{7}{2} \right\rfloor}_{=3} \qquad \text{and} \qquad x_1 \geq \underbrace{\left\lfloor \frac{7}{2} \right\rfloor}_{=4} + 1 \,.$$

Example: execution A, node 2 (1/6)

Node 2: the down-branching child ($x_1 \leq 3$).

- The branching constraint $x_1 \leq 3$ is written as $x_1 + x_5 = 3$, with the slack variable $x_5 \geq 0$
- Using the row of x_1 in the optimal dictionary of LP_1, we have:

$$x_5 = 3 - x_1 = 3 - \left(\frac{7}{2} - \frac{7}{62}\, x_3 + \frac{1}{31}\, x_4\right)$$

$$= -\frac{1}{2} + \frac{7}{62}\, x_3 - \frac{1}{31}\, x_4.$$

- We add this row to the optimal dictionary of LP_1. The reduced costs do not change, so the new dictionary is **dual feasible**; it is not primal feasible, since $\bar{b}_5 = -\frac{1}{2} < 0$

Example: execution A, node 2 (2/6)

- The optimal dictionary of LP_1 with the new row of x_5:

$$\begin{cases} x_1 = \frac{7}{2} - \frac{7}{62} x_3 + \frac{1}{31} x_4 \\ x_2 = \frac{5}{2} + \frac{1}{31} x_3 - \frac{5}{62} x_4 \\ x_5 = -\frac{1}{2} + \frac{7}{62} x_3 - \frac{1}{31} x_4 \\ z = \frac{31}{2} - \frac{17}{62} x_3 - \frac{2}{31} x_4 \end{cases}$$

Example: execution A, node 2 (3/6)

- **Dual pivoting operation.** The pivot row is x_5, the only basic variable with a negative value ($\bar{b}_5 = -\frac{1}{2}$)
- The candidates for the pivot column are the non-basic variables with $\bar{a}_{5,j} < 0$:

$$\frac{\bar{c}_3}{\bar{a}_{5,3}} = \frac{-\frac{17}{62}}{-\frac{7}{62}} = \frac{17}{7}$$

- The pivot column is x_3, with $s_m = \frac{17}{7}$, and the objective value becomes

$$\bar{z} + s_m\, \bar{b}_5 = \frac{31}{2} + \frac{17}{7} \cdot \left(-\frac{1}{2}\right) = \frac{100}{7}.$$

Example: execution A, node 2 (4/6)

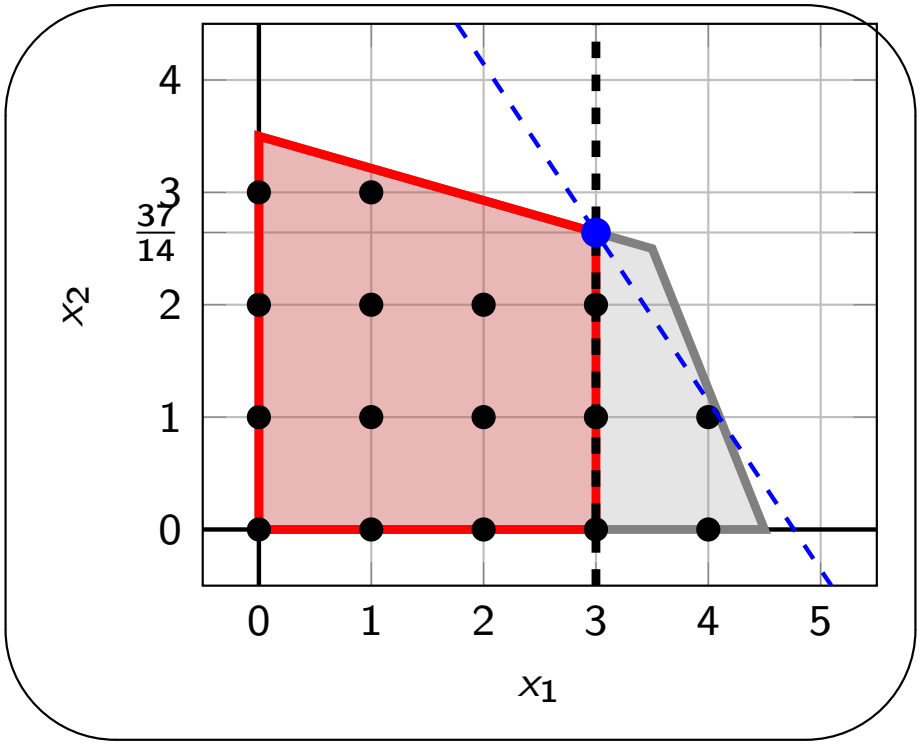
- After the pivoting operation we obtain the following dictionary, which is primal and dual feasible, hence **optimal**:

$$\begin{cases} x_1 = 3 - x_5 \\ x_2 = \frac{37}{14} - \frac{1}{14} x_4 + \frac{2}{7} x_5 \\ x_3 = \frac{31}{7} + \frac{2}{7} x_4 + \frac{62}{7} x_5 \\ z = \frac{100}{7} - \frac{1}{7} x_4 - \frac{17}{7} x_5 \end{cases}$$

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_2)$ is:

$$\tilde{x}_1 = 3, \quad \tilde{x}_2 = \frac{37}{14}, \qquad z(\mathrm{LP}_2) = \frac{100}{7}.$$

Example: execution A, node 2 (5/6)



Node 2: the feasible region of LP_2 (red) inside the region of its parent LP_1 (gray), and the optimal solution of LP_2 (blue) with the objective line through it (dashed).

Example: execution A, node 2 (6/6)

- The solution is **fractional** and

$$\lfloor z(\mathrm{LP}_2) \rfloor = \left\lfloor \frac{100}{7} \right\rfloor = 14 > \mathrm{LB} = -\infty,$$

  so the LP_2 problem is **inserted in the list** L

Example: execution A, node 3 (1/6)

Node 3: the up-branching child ($x_1 \geq 4$).

- The branching constraint $x_1 \geq 4$ is written as $x_1 - x_5 = 4$, with the surplus variable $x_5 \geq 0$
- Using the row of x_1 in the optimal dictionary of LP_1, we have:

$$x_5 = x_1 - 4 = \left(\frac{7}{2} - \frac{7}{62}\, x_3 + \frac{1}{31}\, x_4\right) - 4$$

$$= -\frac{1}{2} - \frac{7}{62}\, x_3 + \frac{1}{31}\, x_4.$$

- We add this row to the optimal dictionary of LP_1. The reduced costs do not change, so the new dictionary is **dual feasible**; it is not primal feasible, since $\bar{b}_5 = -\frac{1}{2} < 0$

Example: execution A, node 3 (2/6)

- The optimal dictionary of LP_1 with the new row of x_5:

$$\begin{cases} x_1 = \frac{7}{2} - \frac{7}{62} x_3 + \frac{1}{31} x_4 \\ x_2 = \frac{5}{2} + \frac{1}{31} x_3 - \frac{5}{62} x_4 \\ x_5 = -\frac{1}{2} - \frac{7}{62} x_3 + \frac{1}{31} x_4 \\ z = \frac{31}{2} - \frac{17}{62} x_3 - \frac{2}{31} x_4 \end{cases}$$

Example: execution A, node 3 (3/6)

- **Dual pivoting operation.** The pivot row is x_5, the only basic variable with a negative value ($\bar{b}_5 = -\frac{1}{2}$)
- The candidates for the pivot column are the non-basic variables with $\bar{a}_{5,j} < 0$:

$$\frac{\bar{c}_4}{\bar{a}_{5,4}} = \frac{-\frac{2}{31}}{-\frac{1}{31}} = 2$$

- The pivot column is x_4, with $s_m = 2$, and the objective value becomes

$$\bar{z} + s_m \bar{b}_5 = \frac{31}{2} + 2 \cdot \left(-\frac{1}{2}\right) = \frac{29}{2}.$$

Example: execution A, node 3 (4/6)

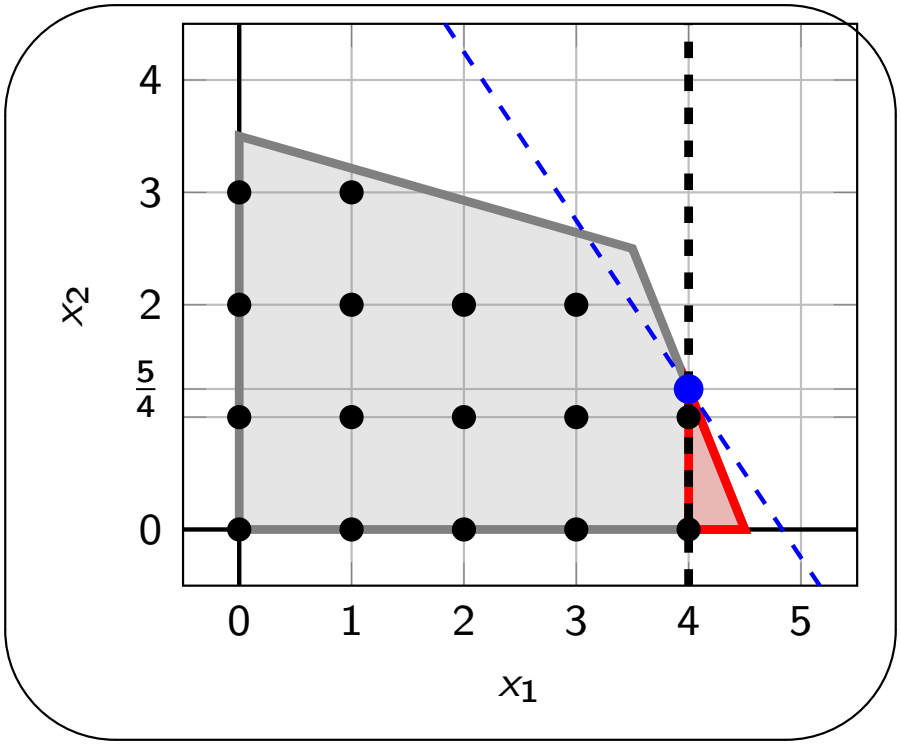
- After the pivoting operation we obtain the following dictionary, which is primal and dual feasible, hence **optimal**:

$$\begin{cases} x_1 = 4 + x_5 \\ x_2 = \frac{5}{4} - \frac{1}{4} x_3 - \frac{5}{2} x_5 \\ x_4 = \frac{31}{2} + \frac{7}{2} x_3 + 31 x_5 \\ z = \frac{29}{2} - \frac{1}{2} x_3 - 2 x_5 \end{cases}$$

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_3)$ is:

$$\tilde{x}_1 = 4, \quad \tilde{x}_2 = \frac{5}{4}, \qquad z(\mathrm{LP}_3) = \frac{29}{2}.$$

Example: execution A, node 3 (5/6)



Node 3: the feasible region of LP_3 (red) inside the region of its parent LP_1 (gray), and the optimal solution of LP_3 (blue) with the objective line through it (dashed).

Example: execution A, node 3 (6/6)

- The solution is **fractional** and

$$\lfloor z(\mathrm{LP}_3) \rfloor = \left\lfloor \frac{29}{2} \right\rfloor = 14 > \mathrm{LB} = -\infty,$$

  so the LP_3 problem is **inserted in the list** L

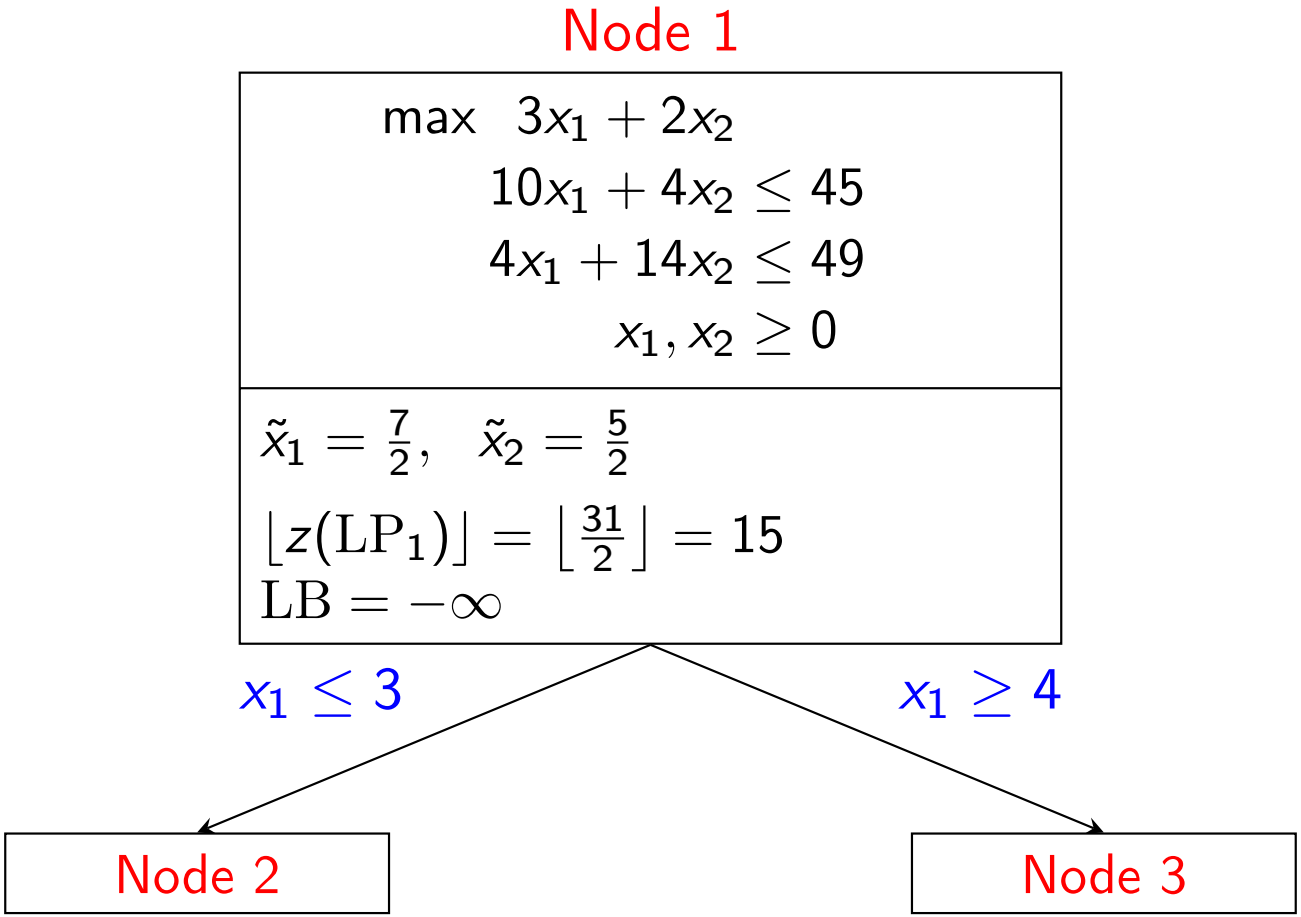
Example: execution A, bounds after iteration 1

- The list is $L = \{\mathrm{LP}_2,\ \mathrm{LP}_3\}$, and the upper bound is the largest bound of the nodes in the list:

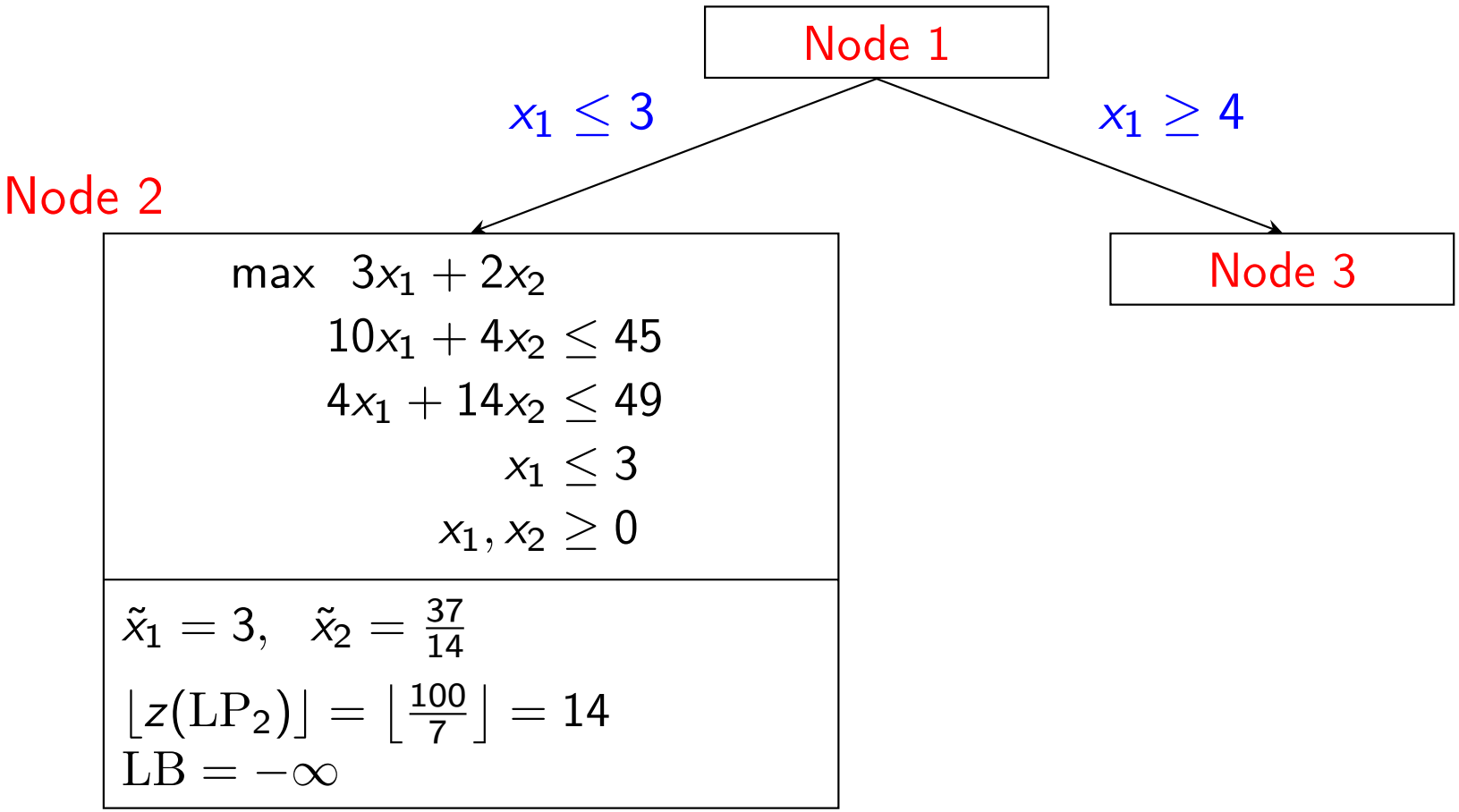
$$\underbrace{-\infty}_{\mathrm{LB}} \ \leq\ z(\mathrm{ILP})\ \leq\ \underbrace{14}_{\mathrm{UB}} \quad \left(\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k) = 14\right).$$

- On the next slides: the portion of the branch-and-bound tree built in the first iteration, with the LP problems of the branching nodes

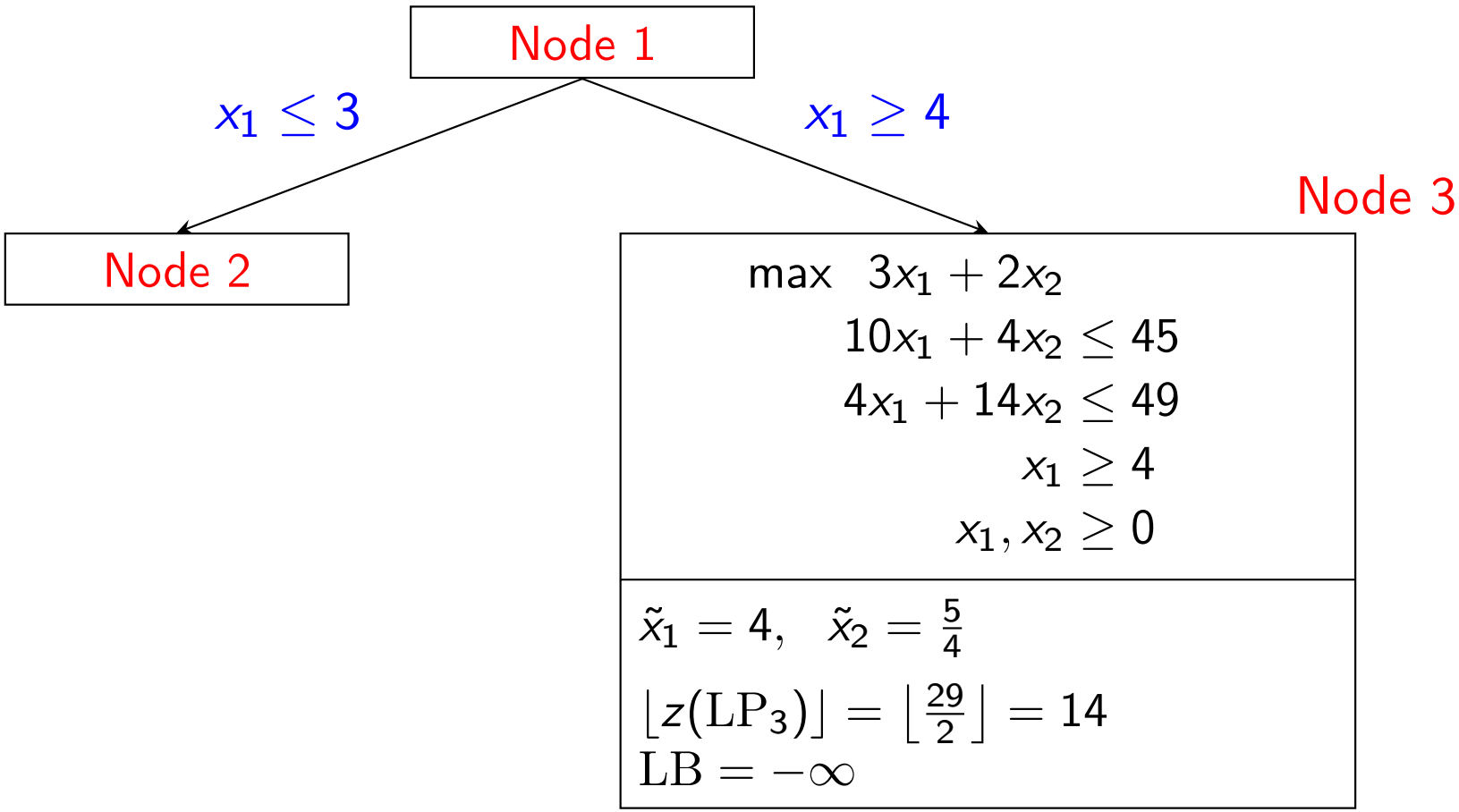
Example: execution A, the tree after iteration 1 (1/3)



Example: execution A, the tree after iteration 1 (2/3)



Example: execution A, the tree after iteration 1 (3/3)



Example: execution A, iteration 2

- The list is $L = \{\mathrm{LP}_2,\ \mathrm{LP}_3\}$; following the LIFO rule, LP_3 (the last problem inserted) is extracted and removed from L
- In $\tilde{\boldsymbol{x}}(\mathrm{LP}_3)$ the only integer variable with a fractional value is x_2, which is the branching variable
- The branching constraints are:

$$x_2 \leq \underbrace{\left\lfloor \frac{5}{4} \right\rfloor}_{=1} \qquad \text{and} \qquad x_2 \geq \underbrace{\left\lfloor \frac{5}{4} \right\rfloor + 1}_{=2}.$$

Example: execution A, node 4 (1/6)

Node 4: the down-branching child ($x_2 \leq 1$).

- The branching constraint $x_2 \leq 1$ is written as $x_2 + x_6 = 1$, with the slack variable $x_6 \geq 0$
- Using the row of x_2 in the optimal dictionary of LP_3, we have:

$$x_6 = 1 - x_2 = 1 - \left(\frac{5}{4} - \frac{1}{4}\,x_3 - \frac{5}{2}\,x_5\right)$$
$$= -\frac{1}{4} + \frac{1}{4}\,x_3 + \frac{5}{2}\,x_5.$$

- We add this row to the optimal dictionary of LP_3. The reduced costs do not change, so the new dictionary is **dual feasible**; it is not primal feasible, since $\bar{b}_6 = -\frac{1}{4} < 0$

Example: execution A, node 4 (2/6)

- The optimal dictionary of LP_3 with the new row of x_6:

$$\begin{cases} x_1 = 4 + x_5 \\ x_2 = \frac{5}{4} - \frac{1}{4} x_3 - \frac{5}{2} x_5 \\ x_4 = \frac{31}{2} + \frac{7}{2} x_3 + 31 x_5 \\ x_6 = -\frac{1}{4} + \frac{1}{4} x_3 + \frac{5}{2} x_5 \\ z = \frac{29}{2} - \frac{1}{2} x_3 - 2 x_5 \end{cases}$$

Example: execution A, node 4 (3/6)

- **Dual pivoting operation.** The pivot row is x_6, the only basic variable with a negative value ($\bar{b}_6 = -\frac{1}{4}$)
- The candidates for the pivot column are the non-basic variables with $\bar{a}_{6,j} < 0$:

$$\frac{\bar{c}_3}{\bar{a}_{6,3}} = \frac{-\frac{1}{2}}{-\frac{1}{4}} = 2, \quad \frac{\bar{c}_5}{\bar{a}_{6,5}} = \frac{-2}{-\frac{5}{2}} = \frac{4}{5} \qquad s_m = \min\left\{2, \frac{4}{5}\right\} = \frac{4}{5}.$$

- The pivot column is x_5, with $s_m = \frac{4}{5}$, and the objective value becomes

$$\bar{z} + s_m\, \bar{b}_6 = \frac{29}{2} + \frac{4}{5} \cdot \left(-\frac{1}{4}\right) = \frac{143}{10}.$$

Example: execution A, node 4 (4/6)

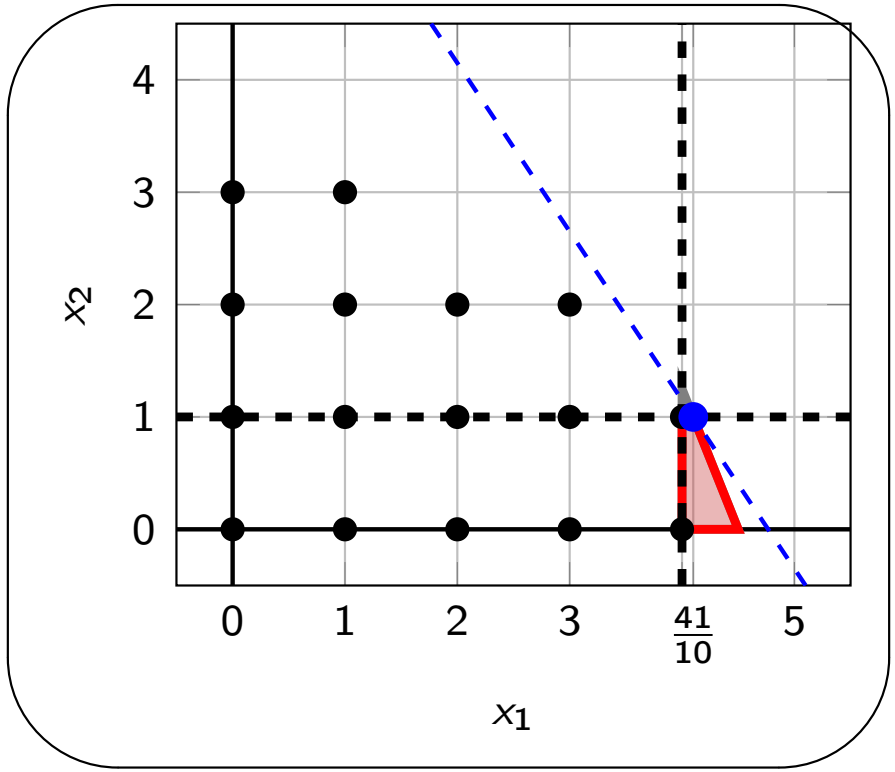
- After the pivoting operation we obtain the following dictionary, which is primal and dual feasible, hence **optimal**:

$$\begin{cases} x_1 = \frac{41}{10} - \frac{1}{10} x_3 + \frac{2}{5} x_6 \\ x_2 = 1 - x_6 \\ x_4 = \frac{93}{5} + \frac{2}{5} x_3 + \frac{62}{5} x_6 \\ x_5 = \frac{1}{10} - \frac{1}{10} x_3 + \frac{2}{5} x_6 \\ z = \frac{143}{10} - \frac{3}{10} x_3 - \frac{4}{5} x_6 \end{cases}$$

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_4)$ is:

$$\tilde{x}_1 = \frac{41}{10}, \quad \tilde{x}_2 = 1, \qquad z(\mathrm{LP}_4) = \frac{143}{10}.$$

Example: execution A, node 4 (5/6)



Node 4: the feasible region of LP_4 (red) inside the region of its parent LP_3 (gray), and the optimal solution of LP_4 (blue) with the objective line through it (dashed).

Example: execution A, node 4 (6/6)

- The solution is **fractional** and

$$\lfloor z(\mathrm{LP}_4) \rfloor = \left\lfloor \frac{143}{10} \right\rfloor = 14 > \mathrm{LB} = -\infty,$$

  so the LP_4 problem is **inserted in the list** L

Example: execution A, node 5 (1/4)

Node 5: the up-branching child ($x_2 \geq 2$).

- The branching constraint $x_2 \geq 2$ is written as $x_2 - x_6 = 2$, with the surplus variable $x_6 \geq 0$
- Using the row of x_2 in the optimal dictionary of LP_3, we have:

$$x_6 = x_2 - 2 = \left(\frac{5}{4} - \frac{1}{4}\,x_3 - \frac{5}{2}\,x_5\right) - 2$$
$$= -\frac{3}{4} - \frac{1}{4}\,x_3 - \frac{5}{2}\,x_5.$$

- We add this row to the optimal dictionary of LP_3. The reduced costs do not change, so the new dictionary is **dual feasible**; it is not primal feasible, since $\bar{b}_6 = -\frac{3}{4} < 0$

Example: execution A, node 5 (2/4)

- The optimal dictionary of LP_3 with the new row of x_6:

$$\begin{cases} x_1 = 4 + x_5 \\ x_2 = \frac{5}{4} - \frac{1}{4} x_3 - \frac{5}{2} x_5 \\ x_4 = \frac{31}{2} + \frac{7}{2} x_3 + 31 x_5 \\ x_6 = -\frac{3}{4} - \frac{1}{4} x_3 - \frac{5}{2} x_5 \\ z = \frac{29}{2} - \frac{1}{2} x_3 - 2 x_5 \end{cases}$$

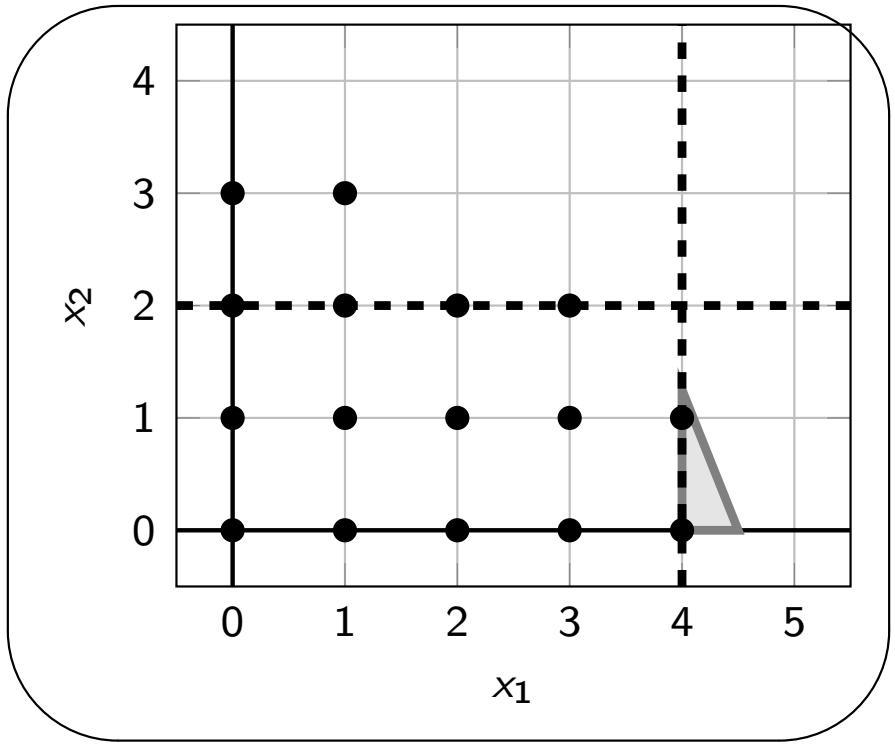
Example: execution A, node 5 (3/4)

- **Dual pivoting operation.** The pivot row is x_6, the only basic variable with a negative value ($\bar{b}_6 = -\frac{3}{4}$)
- All the coefficients of the pivot row are non-negative ($\bar{a}_{6,3} = \frac{1}{4}$, $\bar{a}_{6,5} = \frac{5}{2}$), that is, the row reads

$$x_6 = -\frac{3}{4} - \frac{1}{4}x_3 - \frac{5}{2}x_5 \leq -\frac{3}{4} < 0$$

  for every non-negative value of the non-basic variables
- By the dual simplex algorithm, the LP_5 problem is **infeasible**

Example: execution A, node 5 (4/4)



Node 5: the constraint $x_2 \geq 2$ leaves no point of the region of its parent LP_3 (gray): LP_5 is infeasible.

- Accordingly, Node 5 is pruned by infeasibility

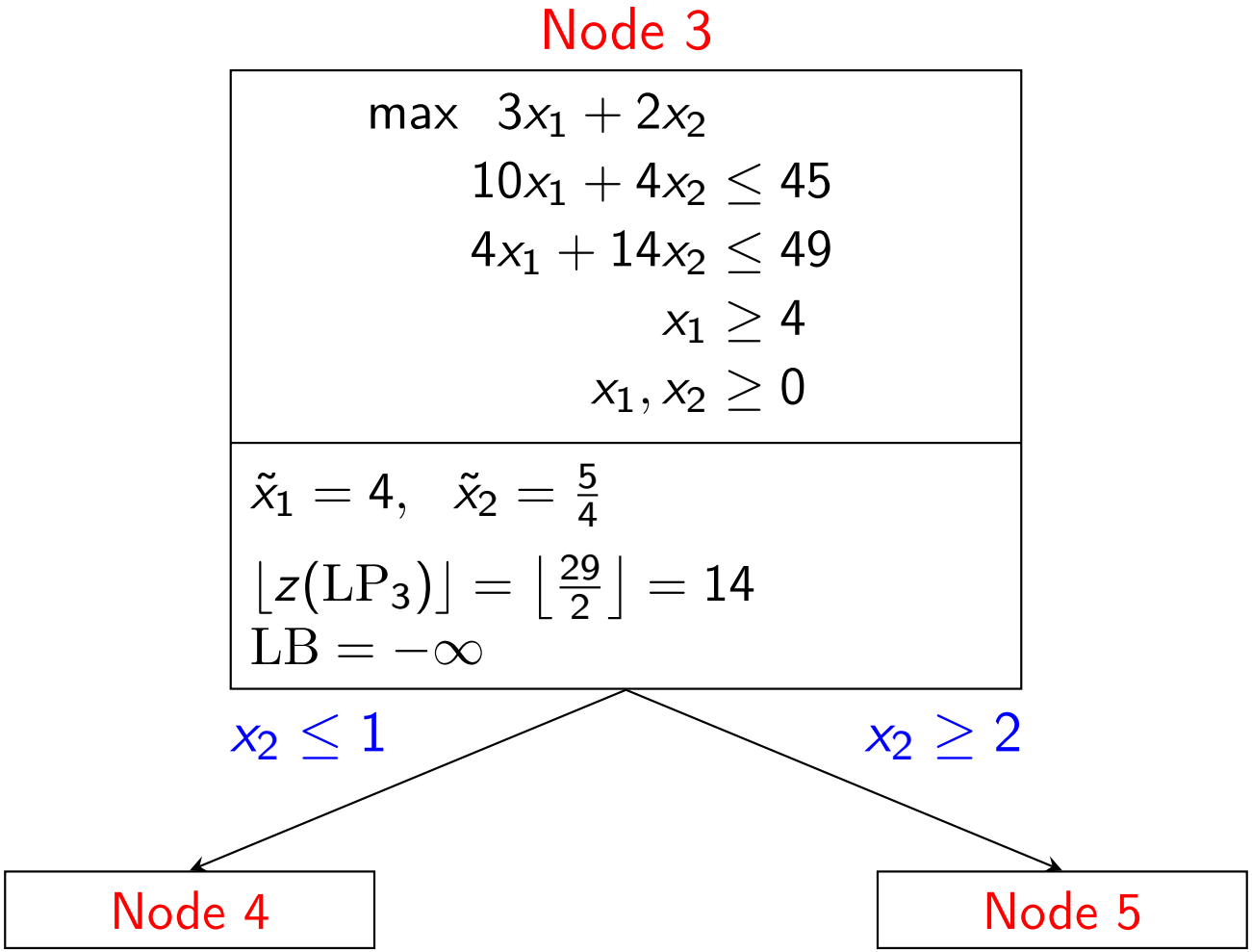
Example: execution A, bounds after iteration 2

- The list is $L = \{\mathrm{LP}_2,\ \mathrm{LP}_4\}$, and the upper bound is the largest bound of the nodes in the list:

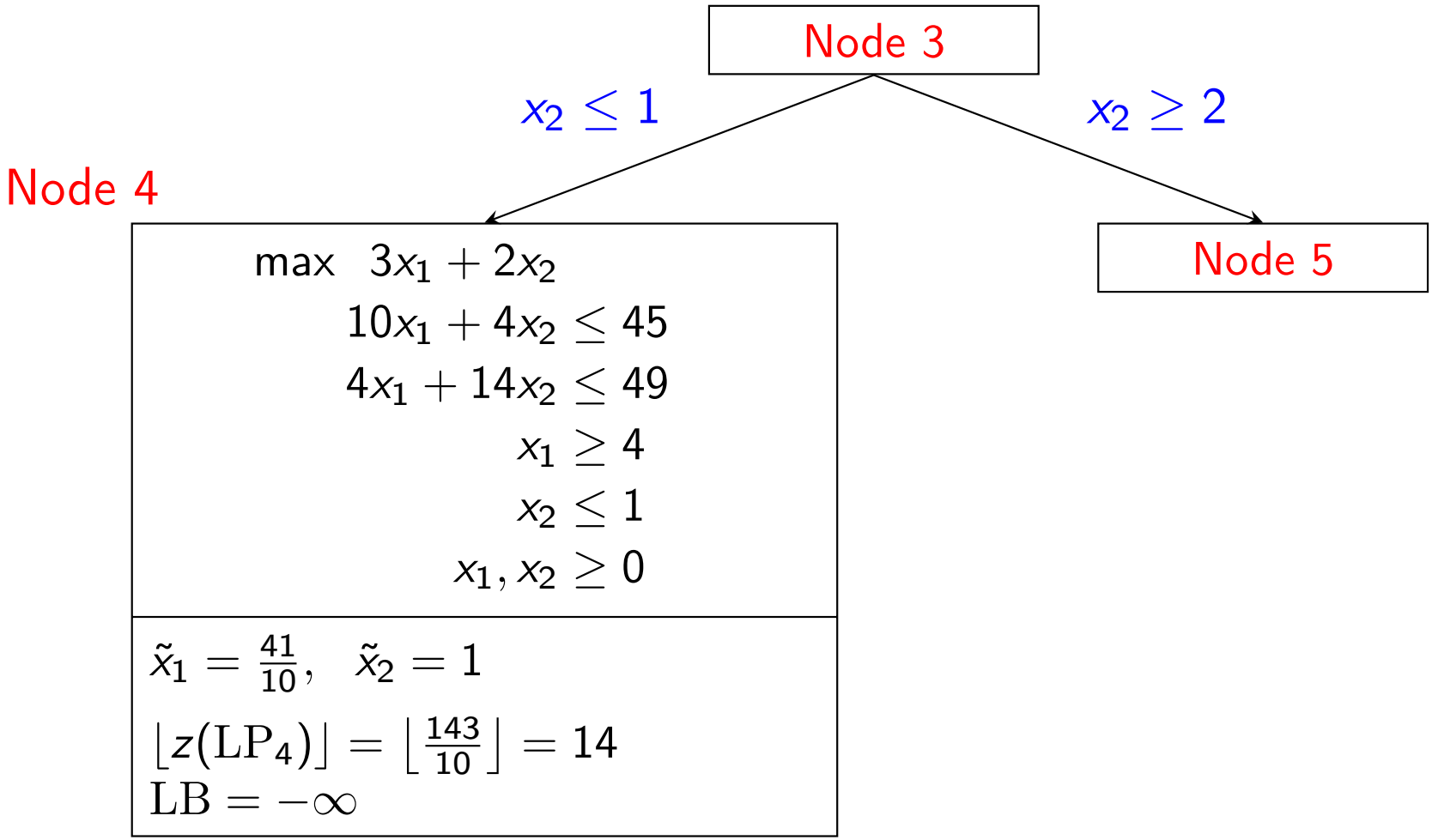
$$\underbrace{-\infty}_{\mathrm{LB}} \ \leq\ z(\mathrm{ILP})\ \leq\ \underbrace{14}_{\mathrm{UB}} \quad \left(\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k) = 14\right).$$

- On the next slides: the portion of the branch-and-bound tree built in the second iteration, with the LP problems of the branching nodes

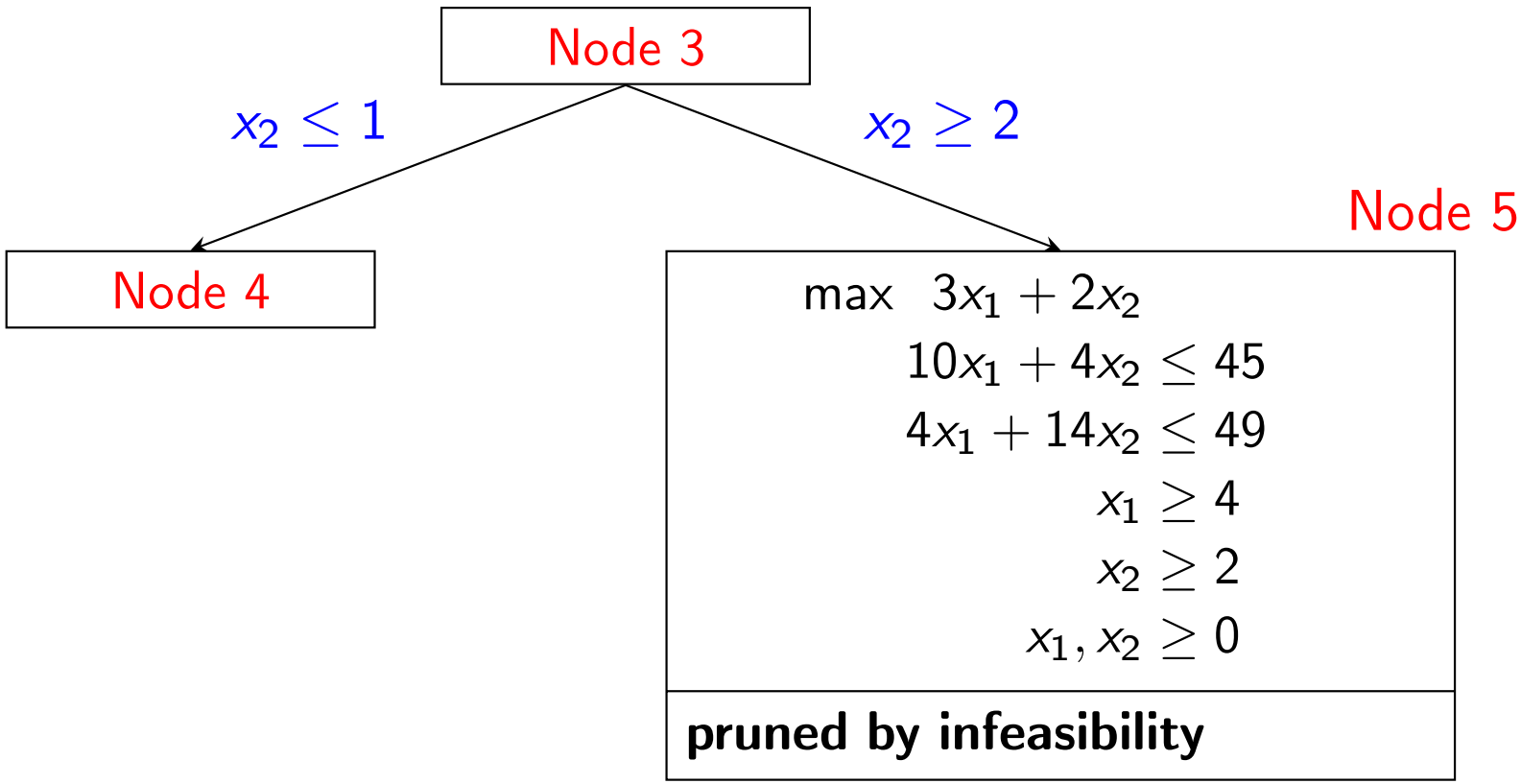
Example: execution A, the tree after iteration 2 (1/3)



Example: execution A, the tree after iteration 2 (2/3)



Example: execution A, the tree after iteration 2 (3/3)



Example: execution A, iteration 3

- The list is $L = \{\mathrm{LP}_2,\ \mathrm{LP}_4\}$; following the LIFO rule, LP_4 (the last problem inserted) is extracted and removed from L
- In $\tilde{\boldsymbol{x}}(\mathrm{LP}_4)$ the only integer variable with a fractional value is x_1, which is the branching variable
- The branching constraints are:

$$x_1 \leq \underbrace{\left\lfloor \frac{41}{10} \right\rfloor}_{=4} \qquad \text{and} \qquad x_1 \geq \underbrace{\left\lfloor \frac{41}{10} \right\rfloor + 1}_{=5}.$$

Example: execution A, node 6 (1/6)

Node 6: the down-branching child ($x_1 \leq 4$).

- The branching constraint $x_1 \leq 4$ is written as $x_1 + x_7 = 4$, with the slack variable $x_7 \geq 0$
- Using the row of x_1 in the optimal dictionary of LP_4, we have:

$$x_7 = 4 - x_1 = 4 - \left(\frac{41}{10} - \frac{1}{10}\,x_3 + \frac{2}{5}\,x_6\right)$$
$$= -\frac{1}{10} + \frac{1}{10}\,x_3 - \frac{2}{5}\,x_6.$$

- We add this row to the optimal dictionary of LP_4. The reduced costs do not change, so the new dictionary is **dual feasible**; it is not primal feasible, since $\bar{b}_7 = -\frac{1}{10} < 0$

Example: execution A, node 6 (2/6)

- The optimal dictionary of LP_4 with the new row of x_7:

$$
\begin{cases}
x_1 = \frac{41}{10} - \frac{1}{10} x_3 + \frac{2}{5} x_6 \\
x_2 = 1 - x_6 \\
x_4 = \frac{93}{5} + \frac{2}{5} x_3 + \frac{62}{5} x_6 \\
x_5 = \frac{1}{10} - \frac{1}{10} x_3 + \frac{2}{5} x_6 \\
x_7 = -\frac{1}{10} + \frac{1}{10} x_3 - \frac{2}{5} x_6 \\
z = \frac{143}{10} - \frac{3}{10} x_3 - \frac{4}{5} x_6
\end{cases}
$$

Example: execution A, node 6 (3/6)

- **Dual pivoting operation.** The pivot row is x_7, the only basic variable with a negative value ($\bar{b}_7 = -\frac{1}{10}$)
- The candidates for the pivot column are the non-basic variables with $\bar{a}_{7,j} < 0$:

$$\frac{\bar{c}_3}{\bar{a}_{7,3}} = \frac{-\frac{3}{10}}{-\frac{1}{10}} = 3$$

- The pivot column is x_3, with $s_m = 3$, and the objective value becomes

$$\bar{z} + s_m \, \bar{b}_7 = \frac{143}{10} + 3 \cdot \left(-\frac{1}{10} \right) = 14.$$

Example: execution A, node 6 (4/6)

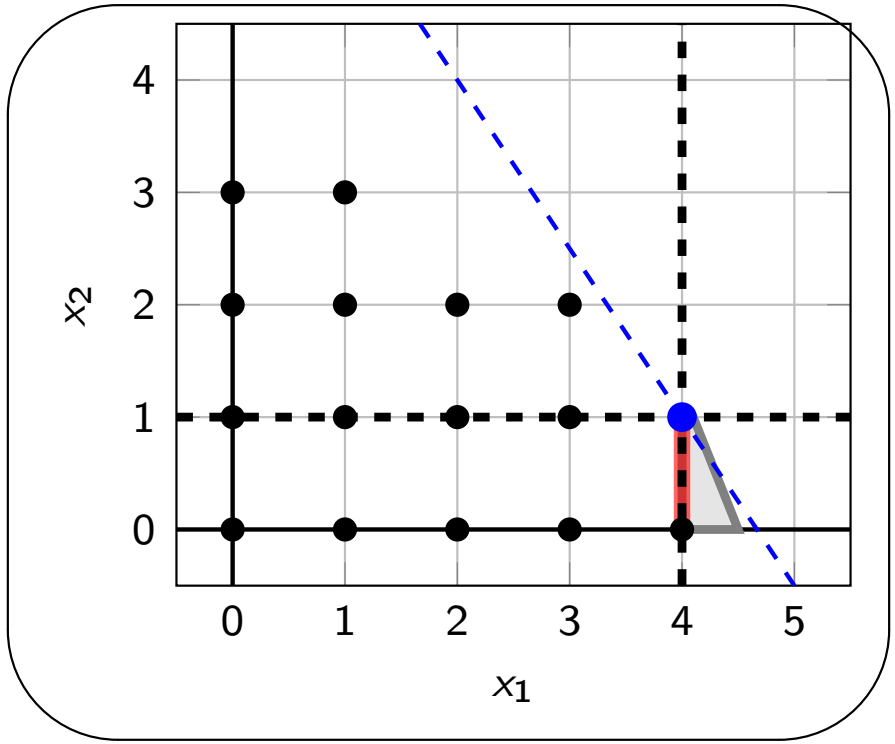
- After the pivoting operation we obtain the following dictionary, which is primal and dual feasible, hence **optimal**:

$$\begin{cases} x_1 = 4 - x_7 \\ x_2 = 1 - x_6 \\ x_3 = 1 + 4x_6 + 10x_7 \\ x_4 = 19 + 14x_6 + 4x_7 \\ x_5 = 0 - x_7 \\ z = 14 - 2x_6 - 3x_7 \end{cases}$$

- An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_6)$ is:

$$\tilde{x}_1 = 4, \quad \tilde{x}_2 = 1, \qquad z(\mathrm{LP}_6) = 14.$$

Example: execution A, node 6 (5/6)



Node 6: the feasible region of LP_6 (red) inside the region of its parent LP_4 (gray), and the optimal solution of LP_6 (blue) with the objective line through it (dashed).

Example: execution A, node 6 (6/6)

- The solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_6)$ is **integer**. Since $z(\mathrm{LP}_6) = 14 > \mathrm{LB} = -\infty$, the lower bound and the incumbent solution are updated:

$$\mathrm{LB} \leftarrow 14, \qquad \bar{\boldsymbol{x}} \leftarrow (4, 1).$$

- Node 6 is pruned by integrality
- Moreover, LP_2 is removed from the list L (pruned by bound), since now

$$\lfloor z(\mathrm{LP}_2) \rfloor = \left\lfloor \frac{100}{7} \right\rfloor = 14 \leq \mathrm{LB} = 14.$$

Example: execution A, the skip rule

- **Skip rule.** After Node 6 the lower bound satisfies

$$\lfloor z(\mathrm{LP}_4) \rfloor = \left\lfloor \frac{143}{10} \right\rfloor = 14 \leq \mathrm{LB} = 14,$$

  so no solution in the region of LP_4 can improve the incumbent solution

- The up-branching child ($x_1 \geq 5$) is **not solved** and it does not receive a node index

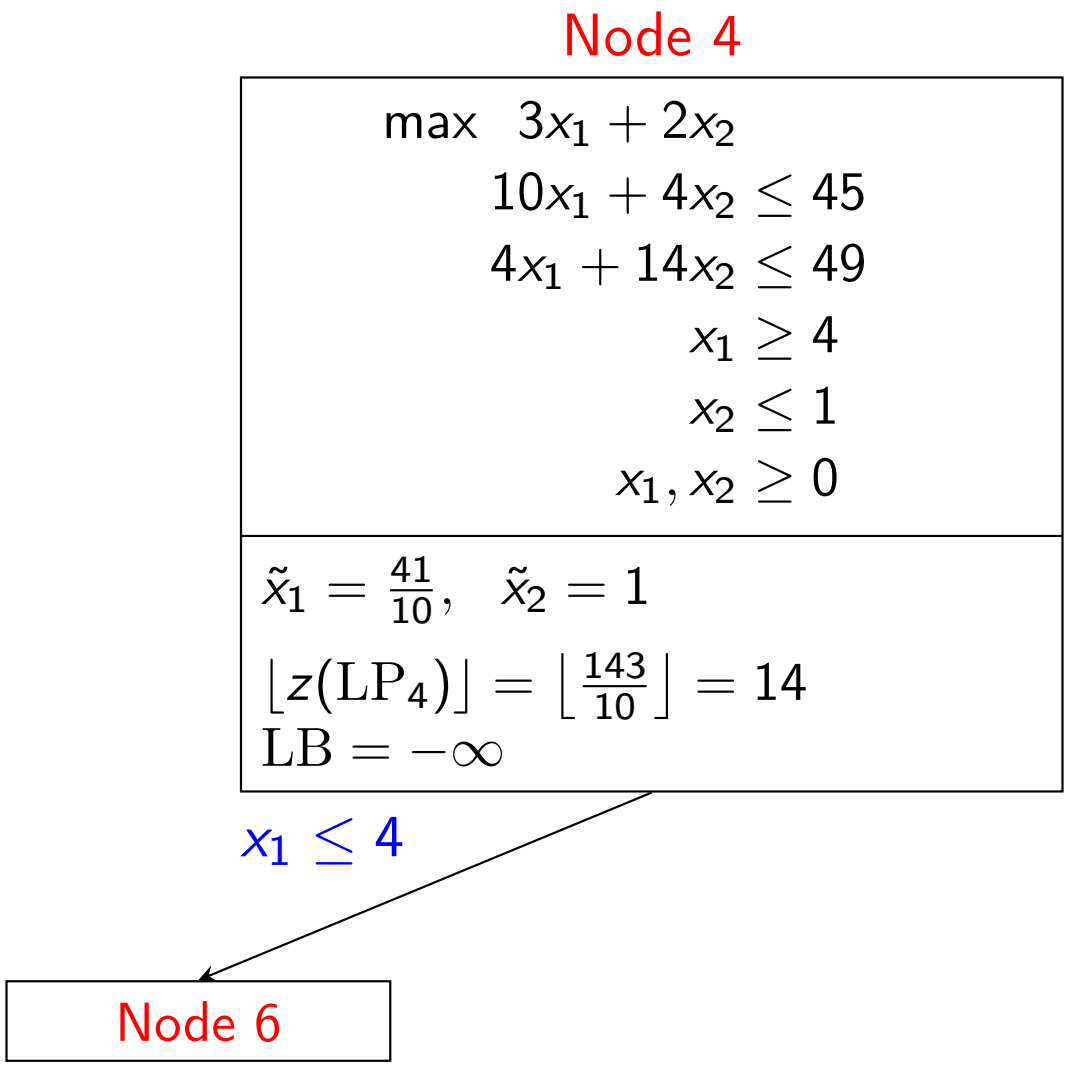
Example: execution A, bounds after iteration 3

- The list is empty, so the upper bound is equal to the lower bound:

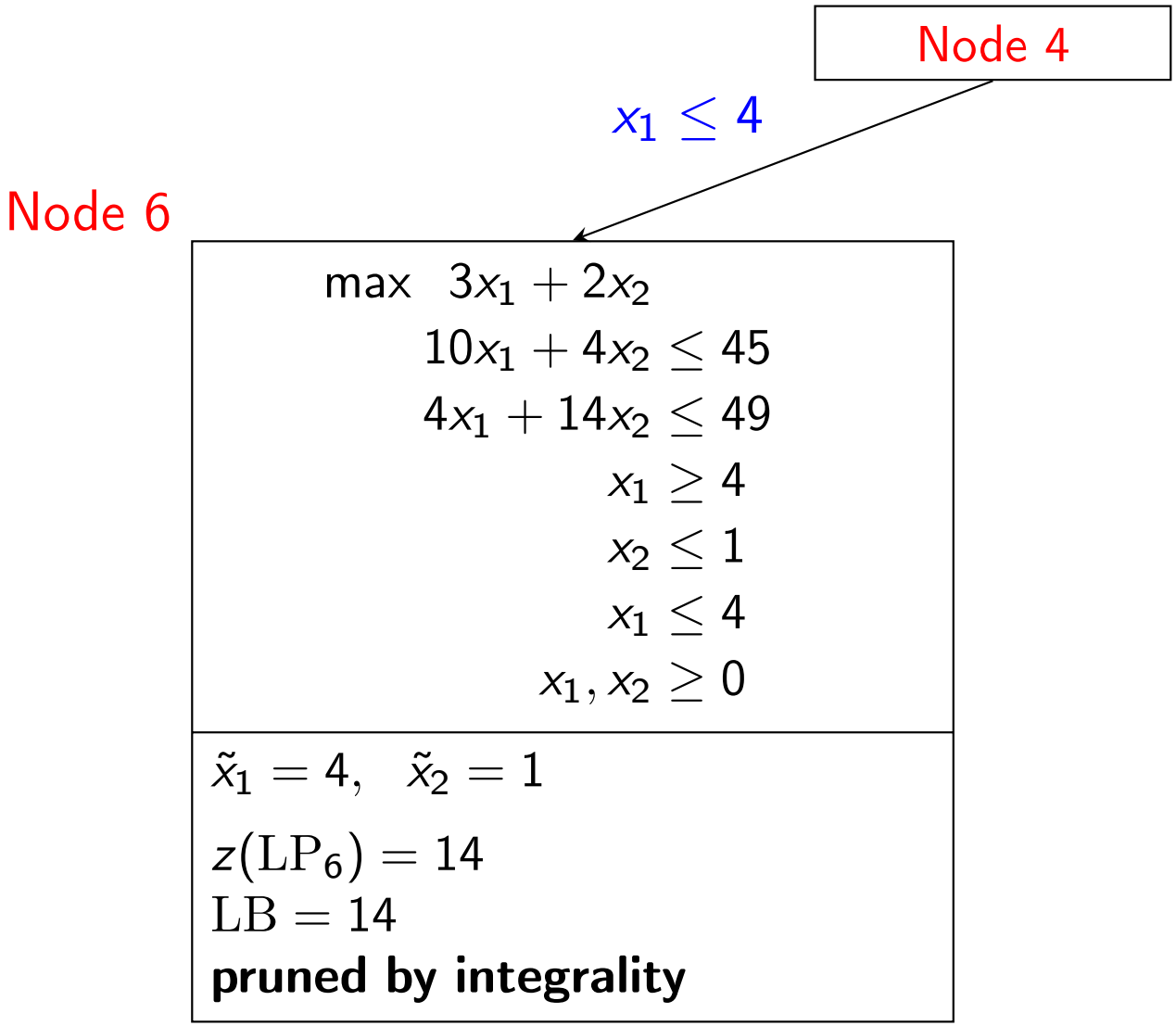
$$\underbrace{14}_{\text{LB}} \leq z(\text{ILP}) \leq \underbrace{14}_{\text{UB}} \quad (\text{UB} = \text{LB} = 14)\,.$$

- On the next slides: the portion of the branch-and-bound tree built in the third iteration, with the LP problems of the branching nodes

Example: execution A, the tree after iteration 3 (1/2)



Example: execution A, the tree after iteration 3 (2/2)



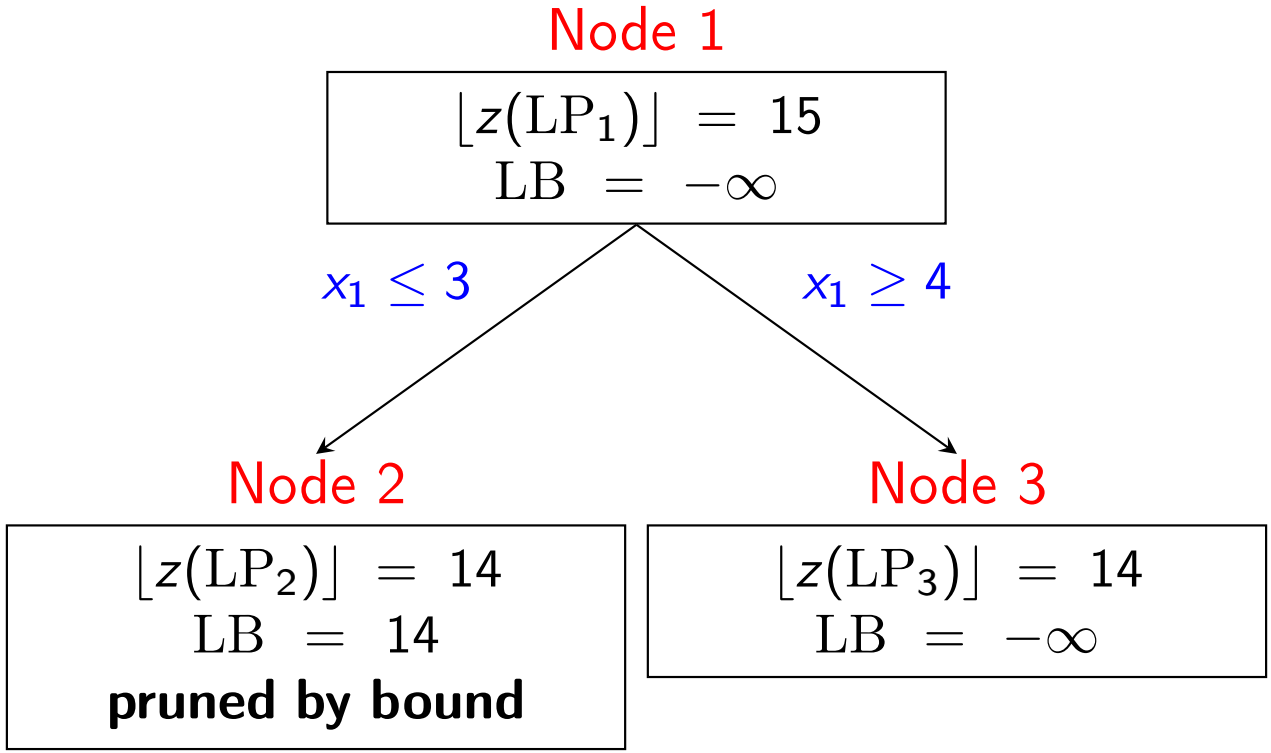
Example: execution A, termination

- The list L is empty: $\mathrm{LB} = \mathrm{UB} = 14$ and the incumbent solution is optimal:

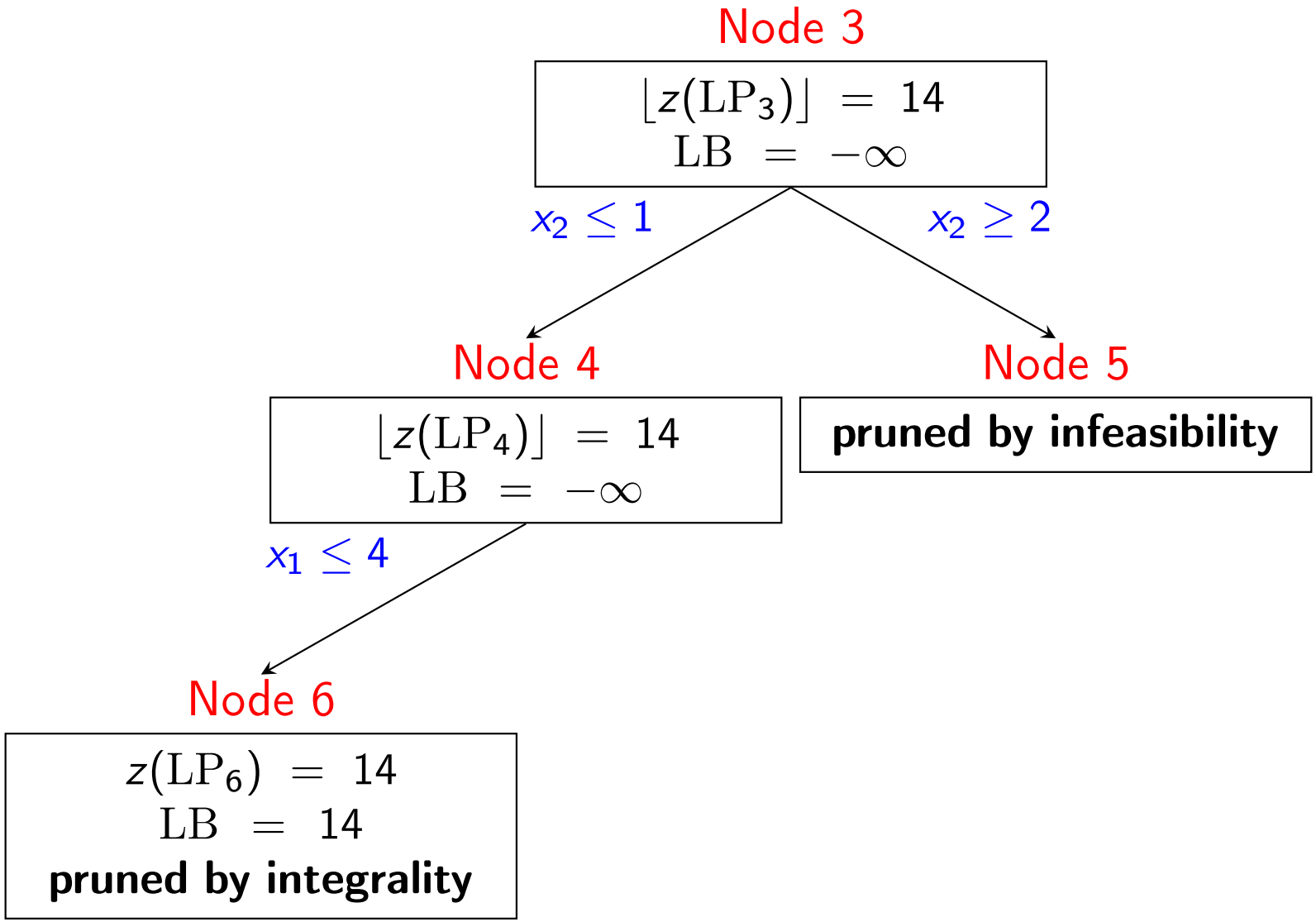
$$\tilde{x}_1 = 4, \quad \tilde{x}_2 = 1, \qquad z(\mathrm{ILP}) = 14.$$

- On the next slides: the final branch-and-bound tree. For each branching node it shows the bound of the node, the lower bound LB when the node was solved (or removed from the list) and the pruning criterion

Example: execution A, the final tree (1/2)



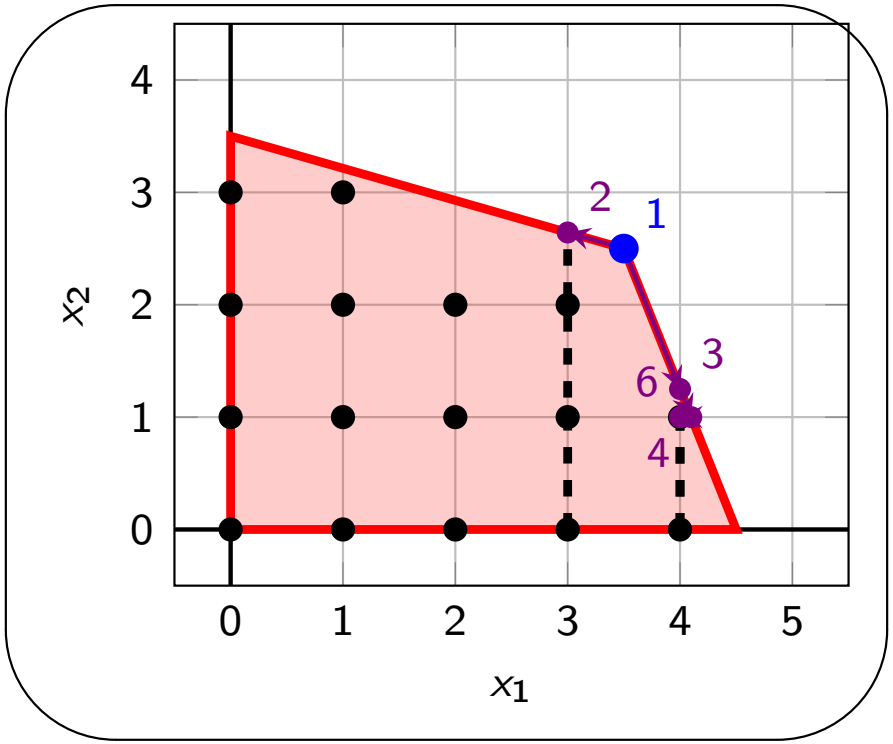
Example: execution A, the final tree (2/2)



Example: execution A in a table

Node	Parent	It.	Branching	$\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$	$z(\mathrm{LP}_k)$	LB	Outcome
1	–	0	–	$(\frac{7}{2}, \frac{5}{2})$	$\frac{31}{2}$	$-\infty$	branched on x_1
2	1	1	$x_1 \leq 3$	$(3, \frac{37}{14})$	$\frac{100}{7}$	14	pruned by bound (removed from L)
3	1	1	$x_1 \geq 4$	$(4, \frac{5}{4})$	$\frac{29}{2}$	$-\infty$	branched on x_2
4	3	2	$x_2 \leq 1$	$(\frac{41}{10}, 1)$	$\frac{143}{10}$	$-\infty$	branched on x_1
5	3	2	$x_2 \geq 2$	–	–	$-\infty$	pruned by infeasibility
6	4	3	$x_1 \leq 4$	$(4, 1)$	14	14	pruned by integrality

Example: execution A in the plane



Execution A in the plane: the branching constraints (black, dashed) inside the regions of the parent nodes, and the optimal solutions of the solved nodes (violet), from LP_1 (blue).

We execute again the B&B algorithm on the car example, changing only the two selection strategies:

the **largest index** and the **best bound**.

Example: execution B, the rules

We execute again the B&B algorithm on the car example, now with:

1. the largest-index rule (lexicographic) as branching-variable selection strategy;
2. the best-bound rule as branching-node selection strategy (ties: smallest node index);
3. the down-branching node is solved first, then the up-branching node;
4. the second child is not solved when the skip rule applies;
5. the lower bound is initialized to $\mathrm{LB} = -\infty$ (no heuristic algorithm is executed).

Each child is re-optimized with the dual simplex algorithm from the optimal dictionary of its parent, exactly as in execution A; here we report only the results.

Example: execution B, initialization

- As before, $\tilde{\boldsymbol{x}}(\mathrm{LP}_1) = (\frac{7}{2}, \frac{5}{2})$ with $z(\mathrm{LP}_1) = \frac{31}{2}$, and LP_1 is inserted in the list L
- The list is $L = \{\mathrm{LP}_1\}$, and the upper bound is the largest bound of the nodes in the list:

$$\underbrace{-\infty}_{\mathrm{LB}} \leq z(\mathrm{ILP}) \leq \underbrace{15}_{\mathrm{UB}} \quad \left(\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k) = 15\right).$$

Example: execution B, iteration 1 (1/3)

- The list is $L = \{\mathrm{LP}_1\}$: the LP_1 problem is extracted and removed from L
- In $\tilde{\boldsymbol{x}}(\mathrm{LP}_1)$ we have $\tilde{x}_1 = \frac{7}{2}$ and $\tilde{x}_2 = \frac{5}{2}$, both fractional; by the largest-index rule the branching variable is x_2
- The branching constraints are:

$$x_2 \leq \underbrace{\left\lfloor \frac{5}{2} \right\rfloor}_{=2} \qquad \text{and} \qquad x_2 \geq \underbrace{\left\lfloor \frac{5}{2} \right\rfloor + 1}_{=3}.$$

Example: execution B, iteration 1 (2/3)

- Node 2 ($x_2 \leq 2$): $\tilde{\boldsymbol{x}}(\mathrm{LP}_2) = (\frac{37}{10}, 2)$ with $z(\mathrm{LP}_2) = \frac{151}{10}$. The solution is **fractional** and

$$\lfloor z(\mathrm{LP}_2) \rfloor = \left\lfloor \frac{151}{10} \right\rfloor = 15 > \mathrm{LB} = -\infty,$$

  so the LP_2 problem is **inserted in the list** L

- Node 3 ($x_2 \geq 3$): $\tilde{\boldsymbol{x}}(\mathrm{LP}_3) = (\frac{7}{4}, 3)$ with $z(\mathrm{LP}_3) = \frac{45}{4}$. The solution is **fractional** and

$$\lfloor z(\mathrm{LP}_3) \rfloor = \left\lfloor \frac{45}{4} \right\rfloor = 11 > \mathrm{LB} = -\infty,$$

  so the LP_3 problem is **inserted in the list** L

Example: execution B, iteration 1 (3/3)

- The list is $L = \{\mathrm{LP}_2,\ \mathrm{LP}_3\}$, and the upper bound is the largest bound of the nodes in the list:

$$\underbrace{-\infty}_{\mathrm{LB}} \ \leq \ z(\mathrm{ILP}) \ \leq \ \underbrace{15}_{\mathrm{UB}} \quad \left(\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k) = 15\right).$$

Example: execution B, iteration 2 (1/4)

- The list is $L = \{\mathrm{LP}_2,\ \mathrm{LP}_3\}$; following the **best-bound** rule, LP_2 is extracted and removed from L, since it has the largest optimal value:

$$z(\mathrm{LP}_2) = \frac{151}{10}, \quad z(\mathrm{LP}_3) = \frac{45}{4}.$$

- In $\tilde{\boldsymbol{x}}(\mathrm{LP}_2)$ the only integer variable with a fractional value is x_1, which is the branching variable

- The branching constraints are:

$$x_1 \leq \underbrace{\left\lfloor \frac{37}{10} \right\rfloor}_{=3} \qquad \text{and} \qquad x_1 \geq \underbrace{\left\lfloor \frac{37}{10} \right\rfloor + 1}_{=4}.$$

Example: execution B, iteration 2 (2/4)

- Node 4 ($x_1 \leq 3$): $\tilde{\boldsymbol{x}}(\mathrm{LP}_4) = (3, 2)$ with $z(\mathrm{LP}_4) = 13$
- The solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_4)$ is **integer**. Since $z(\mathrm{LP}_4) = 13 > \mathrm{LB} = -\infty$, the lower bound and the incumbent solution are updated:

$$\mathrm{LB} \leftarrow 13, \qquad \bar{\boldsymbol{x}} \leftarrow (3, 2).$$

- Node 4 is pruned by integrality
- Moreover, LP_3 is removed from the list L (pruned by bound), since now

$$\lfloor z(\mathrm{LP}_3) \rfloor = \left\lfloor \frac{45}{4} \right\rfloor = 11 \leq \mathrm{LB} = 13.$$

Example: execution B, iteration 2 (3/4)

- Node 5 ($x_1 \geq 4$): $\tilde{\boldsymbol{x}}(\mathrm{LP}_5) = (4, \frac{5}{4})$ with $z(\mathrm{LP}_5) = \frac{29}{2}$
- The solution is **fractional** and

$$\lfloor z(\mathrm{LP}_5) \rfloor = \left\lfloor \frac{29}{2} \right\rfloor = 14 > \mathrm{LB} = 13,$$

so the LP_5 problem is **inserted in the list** L

Example: execution B, iteration 2 (4/4)

- The list is $L = \{\mathrm{LP}_5\}$, and the upper bound is the largest bound of the nodes in the list:

$$\underbrace{13}_{\mathrm{LB}} \leq z(\mathrm{ILP}) \leq \underbrace{14}_{\mathrm{UB}} \quad \left(\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k) = 14\right).$$

Example: execution B, iteration 3 (1/3)

- The list is $L = \{\mathrm{LP}_5\}$: the LP_5 problem is extracted and removed from L
- In $\tilde{\boldsymbol{x}}(\mathrm{LP}_5)$ the only integer variable with a fractional value is x_2, which is the branching variable
- The branching constraints are:

$$x_2 \leq \underbrace{\left\lfloor \frac{5}{4} \right\rfloor}_{=1} \quad \text{and} \quad x_2 \geq \underbrace{\left\lfloor \frac{5}{4} \right\rfloor + 1}_{=2}.$$

Example: execution B, iteration 3 (2/3)

- Node 6 ($x_2 \leq 1$): $\tilde{\boldsymbol{x}}(\mathrm{LP}_6) = (\frac{41}{10}, 1)$ with $z(\mathrm{LP}_6) = \frac{143}{10}$. The solution is **fractional** and

$$\lfloor z(\mathrm{LP}_6) \rfloor = \left\lfloor \frac{143}{10} \right\rfloor = 14 > \mathrm{LB} = 13,$$

  so the LP_6 problem is **inserted in the list** L

- Node 7 ($x_2 \geq 2$): the LP_7 problem is infeasible: pruned by infeasibility

Example: execution B, iteration 3 (3/3)

- The list is $L = \{\mathrm{LP}_6\}$, and the upper bound is the largest bound of the nodes in the list:

$$\underbrace{13}_{\mathrm{LB}} \leq z(\mathrm{ILP}) \leq \underbrace{14}_{\mathrm{UB}} \quad \left(\mathrm{UB} = \max_{k \in L} \mathrm{UB}(\mathrm{LP}_k) = 14\right).$$

Example: execution B, iteration 4 (1/4)

- The list is $L = \{\mathrm{LP}_6\}$: the LP_6 problem is extracted and removed from L
- In $\tilde{\boldsymbol{x}}(\mathrm{LP}_6)$ the only integer variable with a fractional value is x_1, which is the branching variable
- The branching constraints are:

$$x_1 \leq \underbrace{\left\lfloor \frac{41}{10} \right\rfloor}_{=4} \quad \text{and} \quad x_1 \geq \underbrace{\left\lfloor \frac{41}{10} \right\rfloor + 1}_{=5}.$$

Example: execution B, iteration 4 (2/4)

- Node 8 ($x_1 \leq 4$): $\tilde{\boldsymbol{x}}(\mathrm{LP}_8) = (4, 1)$ with $z(\mathrm{LP}_8) = 14$
- The solution $\tilde{\boldsymbol{x}}(\mathrm{LP}_8)$ is **integer**. Since $z(\mathrm{LP}_8) = 14 > \mathrm{LB} = 13$, the lower bound and the incumbent solution are updated:

$$\mathrm{LB} \leftarrow 14, \qquad \bar{\boldsymbol{x}} \leftarrow (4, 1).$$

- Node 8 is pruned by integrality

Example: execution B, iteration 4 (3/4)

- **Skip rule.** After Node 8 the lower bound satisfies

$$\lfloor z(\mathrm{LP}_6) \rfloor = \left\lfloor \frac{143}{10} \right\rfloor = 14 \leq \mathrm{LB} = 14,$$

  so no solution in the region of LP_6 can improve the incumbent solution

- The up-branching child ($x_1 \geq 5$) is **not solved** and it does not receive a node index

Example: execution B, iteration 4 (4/4)

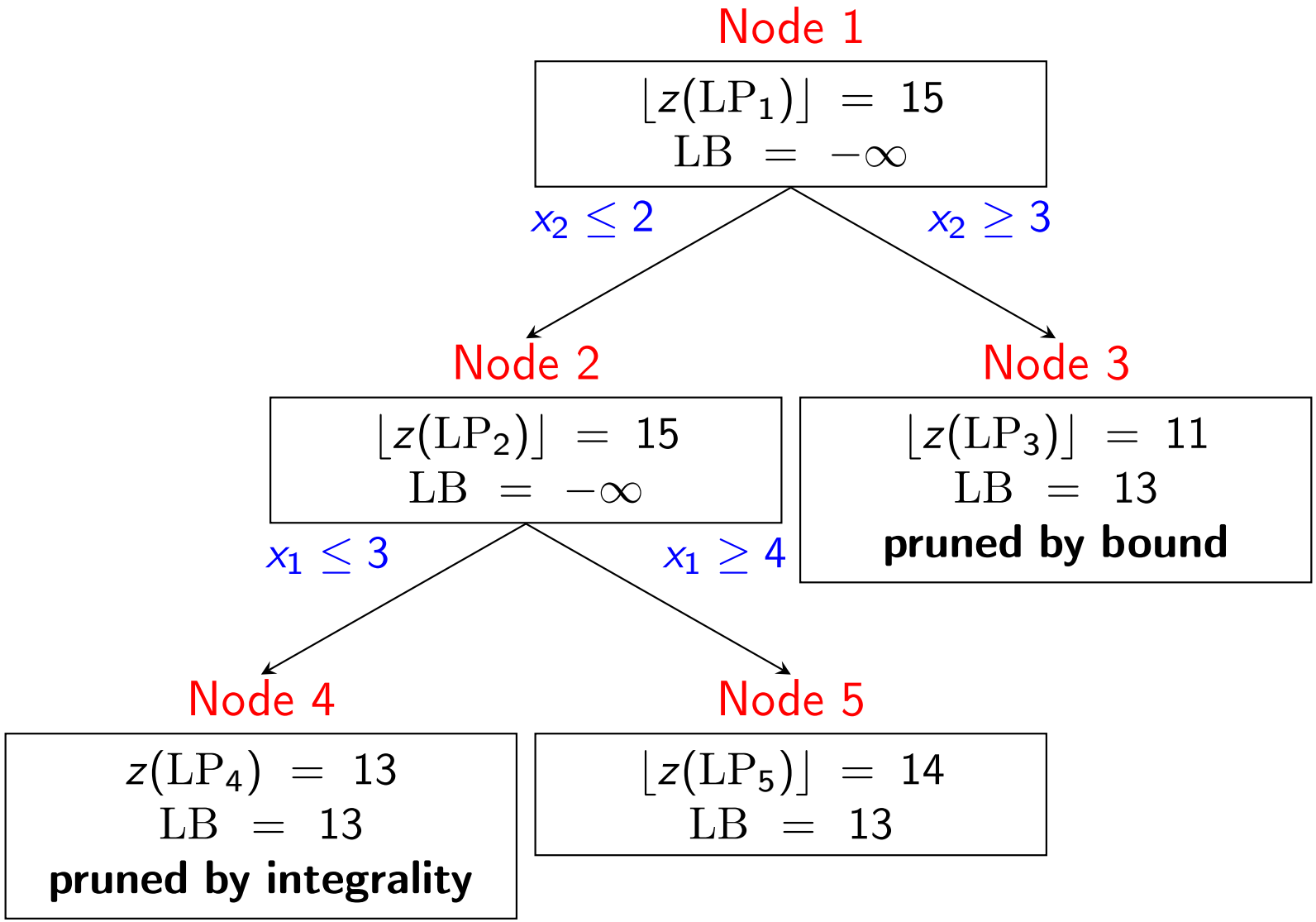
- The list is empty, so the upper bound is equal to the lower bound:

$$\underbrace{14}_{\text{LB}} \leq z(\text{ILP}) \leq \underbrace{14}_{\text{UB}} \quad (\text{UB} = \text{LB} = 14)\,.$$

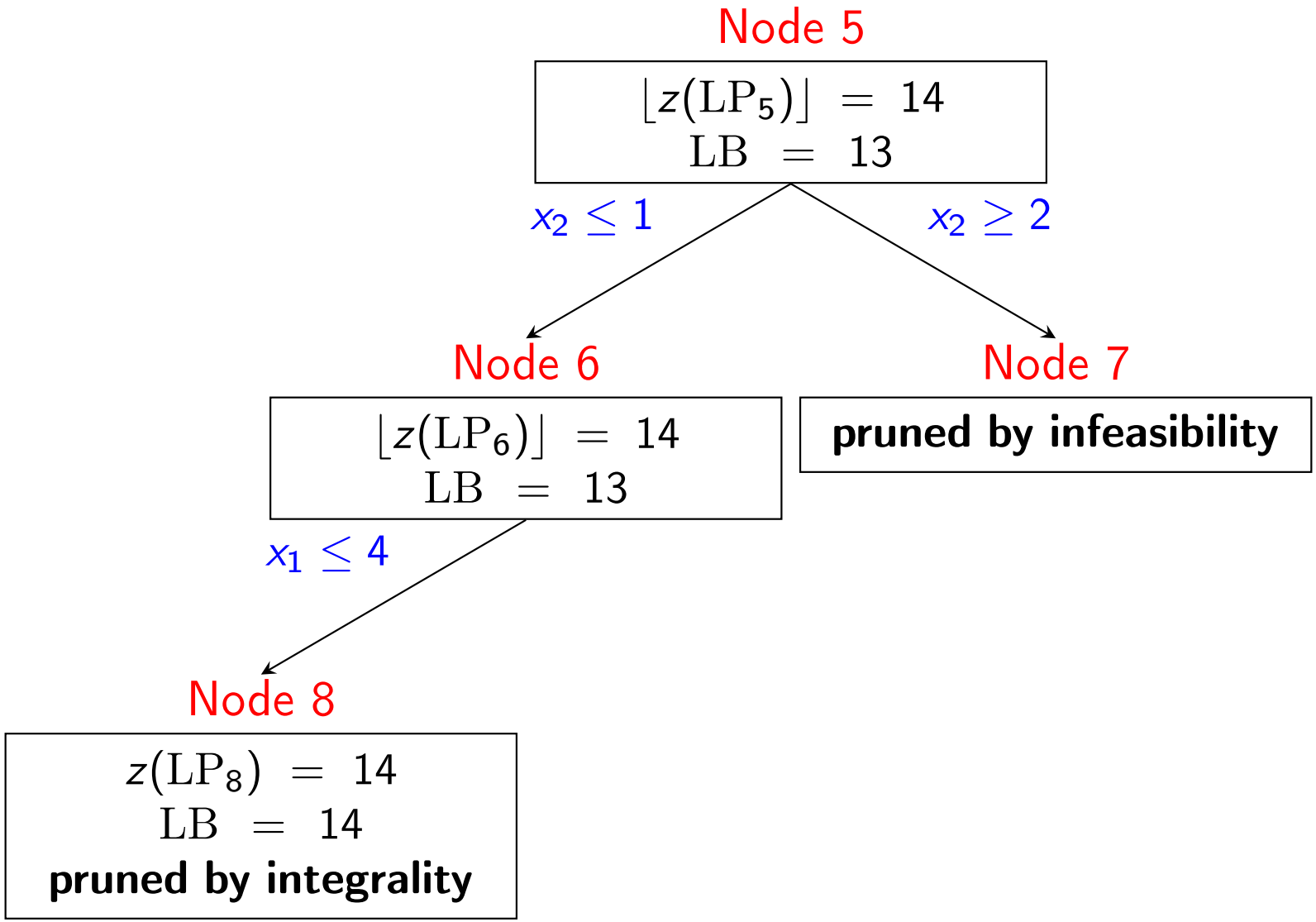
- **Termination.** The list L is empty: $\text{LB} = \text{UB} = 14$ and the incumbent solution is optimal:

$$\tilde{x}_1 = 4, \quad \tilde{x}_2 = 1, \qquad z(\text{ILP}) = 14.$$

Example: execution B, the final tree (1/2)



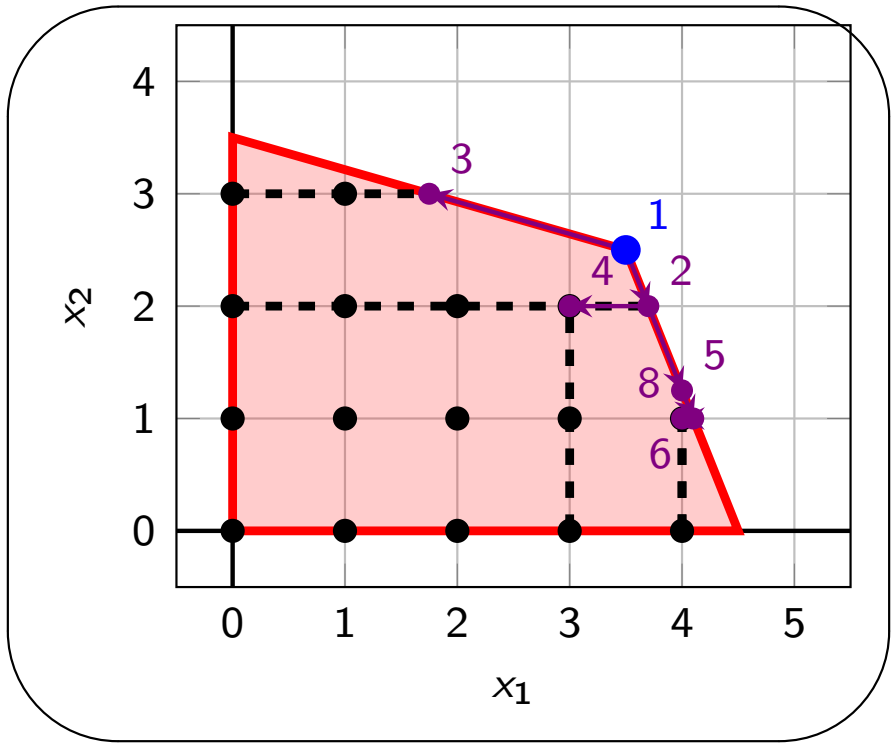
Example: execution B, the final tree (2/2)



Example: execution B in a table

Node	Parent	It.	Branching	$\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$	$z(\mathrm{LP}_k)$	LB	Outcome
1	–	0	–	$(\frac{7}{2}, \frac{5}{2})$	$\frac{31}{2}$	$-\infty$	branched on x_2
2	1	1	$x_2 \leq 2$	$(\frac{37}{10}, 2)$	$\frac{151}{10}$	$-\infty$	branched on x_1
3	1	1	$x_2 \geq 3$	$(\frac{7}{4}, 3)$	$\frac{45}{4}$	13	pruned by bound (removed from L)
4	2	2	$x_1 \leq 3$	$(3, 2)$	13	13	pruned by integrality
5	2	2	$x_1 \geq 4$	$(4, \frac{5}{4})$	$\frac{29}{2}$	13	branched on x_2
6	5	3	$x_2 \leq 1$	$(\frac{41}{10}, 1)$	$\frac{143}{10}$	13	branched on x_1
7	5	3	$x_2 \geq 2$	–	–	13	pruned by infeasibility
8	6	4	$x_1 \leq 4$	$(4, 1)$	14	14	pruned by integrality

Example: execution B in the plane



Execution B in the plane: branching constraints and optimal solutions of the solved nodes.

Example: comparing the two executions

- Execution A solves 6 LP problems, execution B solves 8
- Both certify the same optimal solution $\tilde{x}_1 = 4, \tilde{x}_2 = 1$ of value 14, which is unique
- Execution B finds a first incumbent solution of value 13 earlier, but branching first on x_2 leads to a larger tree on this instance

We execute the B&B algorithm of execution A
(smallest index, LIFO),
starting from a **heuristic solution** of the car example.

Example: heuristic initialization (1/3)

- The one-variable heuristic of the chapter «Primal and dual bounds» fixes x_1 as large as possible and $x_2 = 0$: it gives $\bar{\boldsymbol{x}} = (4, 0)$ of value 12
- Starting execution A with $\mathrm{LB} = 12$, the algorithm solves 6 LP problems, as many as without the heuristic
- This lower bound is too weak to prune any node earlier

Example: heuristic initialization (2/3)

- The heuristic that fixes first x_1 and then x_2 as large as possible gives $\bar{\boldsymbol{x}} = (4, 1)$ of value 14
- Starting with $\mathrm{LB} = 14$, the root is not pruned, since $\lfloor z(\mathrm{LP}_1) \rfloor = 15 > 14$
- Both children are pruned by bound and the algorithm stops after 3 LP problems (table on the next slide)

Example: heuristic initialization (3/3)

Node	Parent	It.	Branching	$\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$	$z(\mathrm{LP}_k)$	LB	Outcome
1	–	0	–	$(\frac{7}{2}, \frac{5}{2})$	$\frac{31}{2}$	14	branched on x_1
2	1	1	$x_1 \leq 3$	$(3, \frac{37}{14})$	$\frac{100}{7}$	14	pruned by bound
3	1	1	$x_1 \geq 4$	$(4, \frac{5}{4})$	$\frac{29}{2}$	14	pruned by bound

- The incumbent solution given by the heuristic is optimal, and the branch-and-bound algorithm **certifies** its optimality

Summary (1/2)

Execution A (smallest index, LIFO):

- Every child is re-optimized from the dictionary of its parent with at most one dual pivoting operation
- 6 LP problems; Node 5 infeasible, Node 6 integer with value 14, Node 2 pruned by bound, one child skipped

Execution B (largest index, best bound):

- 8 LP problems; a first incumbent of value 13, then $(4, 1)$ of value 14

Summary (2/2)

Heuristic initialization:

- $\mathrm{LB} = 12$ prunes nothing earlier: still 6 LP problems
- $\mathrm{LB} = 14$: both children of the root are pruned by bound, 3 LP problems

In all the executions:

- The same certified optimal solution $(4, 1)$, with $z(\mathrm{ILP}) = 14$
- The strategies and the initial lower bound change only the size of the tree

Typical mistakes

Introduction: Typical mistakes

- We list the mistakes that are made most often when executing a B&B algorithm by hand
- They concern rounding, the choice of the branching variable, the bounds, the pruning and the numbering of the nodes
- Knowing them helps to solve the exercises correctly

Observation 1: Typical mistakes (1/3)

- ***Rounding the LP solution.*** *Rounding an optimal solution of the LP relaxation gives at best a feasible solution, in general not an optimal one.*

  In the car example, rounding down $\tilde{\boldsymbol{x}}(\mathrm{LP}) = (\frac{7}{2}, \frac{5}{2})$ *gives* $(3, 2)$ *of value* 13*, while* $z(\mathrm{ILP}) = 14$ *at* $(4, 1)$*.*

Observation: Typical mistakes (2/3)

- ***Branching on the wrong variable.*** *The branching variable must be an integer variable with a fractional value: a variable with an integer value does not cut off* $\tilde{\boldsymbol{x}}(\mathrm{LP}_k)$.
- ***Taking the upper bound from the last node.*** *The upper bound is the largest bound among the nodes of the list L, not the bound of the last node solved.*

Observation: Typical mistakes (3/3)

- ***Pruning by bound only with a strict inequality.*** *A node with* $\mathrm{UB}(\mathrm{LP}_k) = \mathrm{LB}$ *is pruned: its region can contain solutions as good as the incumbent one, but not better.*
- ***Numbering the skipped children.*** *A child that is not solved does not receive a node index.*
- ***Solving every node from scratch.*** *The answer is correct, but much slower: each child starts from the optimal dictionary of its parent, which is dual feasible.*

Summary

Typical mistakes:

- Rounding the LP solution does not give an optimal solution
- Branch only on integer variables with a fractional value
- UB is the largest bound in the list L, not the bound of the last node
- $\mathrm{UB}(\mathrm{LP}_k) = \mathrm{LB}$ is enough to prune by bound
- Skipped children receive no index; children start from the dictionary of the parent

Extensions

Introduction: Extensions

- The algorithm of this chapter can be generalized in two directions
- Another way to solve the LP problems of the nodes, and problems where only some variables are integer

Extensions

- The LP problems of the nodes can also be solved with the **bounded-variable simplex algorithm**, which handles the branching constraints as bounds on the variables instead of new constraints

- In a **mixed** problem, where only some variables are integer, one branches only on the integer variables and the bound of a node is $z(\mathrm{LP}_k)$, without rounding

Summary

- **Bounded-variable simplex algorithm**: branching constraints as bounds on the variables
- **Mixed problems**: branching only on the integer variables, and $\mathrm{UB}(\mathrm{LP}_k) = z(\mathrm{LP}_k)$ without rounding

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- We close with the dashboard of the car example: the bounds found so far in the module

Summary (1/3)

- A B&B algorithm divides the feasible region with branching constraints on integer variables with fractional values, and it never loses a feasible solution
- Every node is pruned by infeasibility, by integrality or by bound, or it waits in the list; the bound of a node is $\lfloor z(\mathrm{LP}_k) \rfloor$ only if all the variables and costs are integer
- The lower bound comes from the incumbent solution, the upper bound from the nodes of the list; when the list is empty they coincide, and the incumbent solution is certified optimal

Summary (2/3)

- Each child is re-optimized with the dual simplex algorithm from the dictionary of its parent
- The selection strategies and a good heuristic solution change the size of the tree, not the optimal value
- In the car example the algorithm closes the gap between $\mathrm{LB} = 13$ (rounded LP solution) and $\mathrm{UB} = 15$ (LP relaxation) by solving 6 LP problems

Summary (3/3)

Dashboard of the car example. The lower and upper bounds on $z(\mathrm{ILP})$ found so far in this module:

Chapter	Lower bound	Upper bound	From
Primal and dual bounds	14	15	heuristic solution $(4, 1)$, and $\lfloor z(\mathrm{LP}) \rfloor$ with $z(\mathrm{LP}) = \frac{31}{2}$
Cutting plane algorithm	14	14	5 Gomory fractional cuts
This chapter, execution A	14	14	6 LP problems solved
This chapter, execution B	14	14	8 LP problems solved
This chapter, with $\mathrm{LB} = 14$	14	14	3 LP problems solved